

Privacy-Focused Menstrual Cycle Tracking System

Capstone 1

Monica Barbosa
Kristian Correa
Gabriel Garcia
Marcelo Hernandez

Dr. Masoud Sadjadi
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1. Problem Statement

Users rely on menstrual cycle tracking applications to monitor reproductive health, fertility, and symptoms. However, many existing applications require users to store highly sensitive personal health data on third-party cloud servers. This raises significant concerns regarding privacy, security, and trust.

These concerns have become more critical in environments where reproductive health data may be legally or socially sensitive, particularly following the overturning of *Roe v. Wade* in 2022. As a result, many users:

- Feel unsafe using cycle tracking applications
- Reduce or completely stop tracking their cycles
- Lose access to valuable health insights and predictive data

In addition to privacy concerns, current solutions often fail to meet modern user expectations, including lack of data ownership, limited customization, and poor inclusivity for irregular cycles.

2. Proposed Solution

This project proposes a privacy-first menstrual cycle tracking system that empowers users with full control over their personal health data. The system uses a local-first architecture, supports user-controlled sharing, provides adaptive predictions for both regular and irregular cycles, and enables deep customization beyond visual themes.

3. Functional Requirements

1. Allow users to record menstrual cycle data.
2. Generate predictions for regular and irregular cycles.
3. Store all data locally by default.
4. Support optional cloud synchronization.
5. Allow controlled data sharing with trusted individuals.
6. Provide customization options and interactive features.

4. Non-Functional Requirements

1. Privacy-first design.
2. Encrypted data storage.
3. Fast and reliable performance.
4. Simple and intuitive UI.
5. Scalable architecture.

5. Summary

This project presents a privacy-focused menstrual cycle tracking system that prioritizes local data storage, user control, and adaptive prediction models. It offers a modern alternative to existing solutions by combining privacy, personalization, and functionality.